# SN74AVCH4T245-Q1 Automotive 4-Bit Dual-Supply Bus Transceiver With Configurable Level-Shifting, Voltage Translation, and 3-State Outputs

## 1 Features

- Control inputs $V_{IH}/V_{IL}$ levels are referenced to $V_{CCA}$ voltage
- Fully configurable dual-rail design allows each port to operate over the full 1.08V to 3.6V power-supply range
- $I_{off}$ supports partial power-down-mode operation
- Bus hold on data inputs eliminates the need for external pull-up/pull-down resistors
- Supports data rate up to:
  - 500Mbps (1.08V to 3.6V translation)
- Latch-up performance exceeds 100mA per JESD 78, class II
- ESD protection exceeds JESD 22:
  - 8000V Human Body Model (A114-A)
  - 200V Machine Model (A115-A)
  - 1000V Charged-Device Model (C101)

## 2 Applications

- Personal electronics
- Industrial
- Enterprise
- Telecom

## 3 Description

This 4-bit noninverting bus transceiver uses two separate configurable power-supply rails. The A port is designed to track $V_{CCA}$. $V_{CCA}$ accepts any supply voltage from 1.08V to 3.6V. The B port is designed to track $V_{CCB}$. $V_{CCB}$ accepts any supply voltage from 1.08V to 3.6V. The SN74AVCH4T245-Q1 is optimized to operate with $V_{CCA}/V_{CCB}$ set at 1.08V to 3.6V. It is operational with $V_{CCA}/V_{CCB}$ as low as 1.08V. This allows for universal low voltage bidirectional translation between any of the 1.2V, 1.5V, 1.8V, 2.5V, and 3.3V voltage nodes.

The SN74AVCH4T245-Q1 is designed for asynchronous communication between two data buses. The logic levels of the direction-control (DIR) input and the output-enable ($\overline{OE}$) input activate either the B-port outputs or the A-port outputs or place both output ports into the high-impedance mode. The device transmits data from the A bus to the B bus when the B-port outputs are activated, and from the B bus to the A bus when the A-port outputs are activated. The input circuitry on both A and B ports is always active and must have a logic HIGH or LOW level applied to prevent excess $I_{CC}$ and $I_{CCZ}$.

The SN74AVCH4T245-Q1 device control pins (1DIR, 2DIR, 1OE, and 2OE) are supplied by $V_{CCA}$.

This device is fully specified for partial-power-down applications using $I_{off}$. The $I_{off}$ circuitry disables the outputs, preventing damaging current backflow through the device when it is powered down.

The $V_{CC}$ isolation feature is designed so that if either $V_{CC}$ input is at GND, then both ports are in the high-impedance state. The bus-hold circuitry on the powered-up side always stays active.

Active bus-hold circuitry holds unused or undriven data inputs at a valid logic state. Use of pull-up or pull-down resistors with the bus-hold circuitry is not recommended. The bus-hold circuitry on the powered-up side always stays active.

To put the device in the high-impedance state during power up or power down, tie the $\overline{OE}$ pin to $V_{CC}$ through a pull-up resistor; the current-sinking capability of the driver determines the minimum value of the resistor.

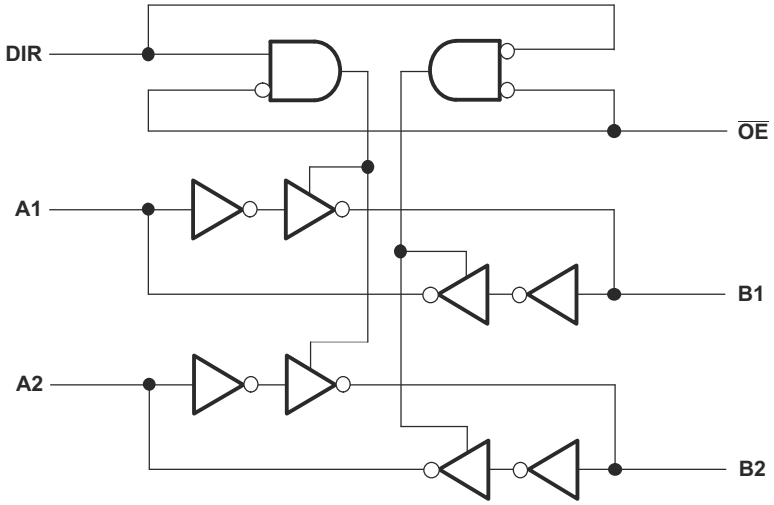
**Package Information**

| PART NUMBER | PACKAGE(1) | PACKAGE SIZE(2) |
|---|---|---|
| SN74AVCH4T245-Q1 | PW (TSSOP, 16) | 5mm × 6.4mm |

(1) For more information, see Section 10.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.

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**Logic Diagram (Positive Logic) for 1/2 of SN74AVCH4T245-Q1**

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# 4 Pin Configuration and Functions

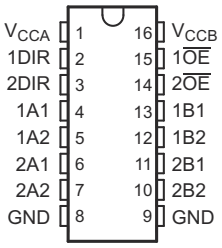



**Figure 4-1. PW Package, 16-Pin TSSOP (Top View)**

**Table 4-1. Pin Functions**

| PIN | | TYPE[1] | DESCRIPTION |
|---|---|---|---|
| NAME | NO. | | |
| 1A1 | 4 | I/O | Input/output 1A1. Referenced to $V_{CCA}$. |
| 1A2 | 5 | I/O | Input/output 1A2. Referenced to $V_{CCA}$. |
| 1B1 | 13 | I/O | Input/output 1B1. Referenced to $V_{CCB}$. |
| 1B2 | 12 | I/O | Input/output 1B2. Referenced to $V_{CCB}$. |
| 1DIR | 2 | I | Direction-control input for 1 ports |
| 1 $\overline{OE}$ | 15 | I | 3-state output-mode enables. Pull $\overline{OE}$ high to place '1' outputs in 3-state mode. Referenced to $V_{CCA}$. |
| 2A1 | 6 | I/O | Input/output 2A1. Referenced to $V_{CCA}$. |
| 2A2 | 7 | I/O | Input/output 2A2. Referenced to $V_{CCA}$. |
| 2B1 | 11 | I/O | Input/output 2B1. Referenced to $V_{CCB}$. |
| 2B2 | 10 | I/O | Input/output 2B2. Referenced to $V_{CCB}$. |
| 2DIR | 3 | I | Direction-control input for 2 ports |
| 2 $\overline{OE}$ | 14 | I | 3-state output-mode enables. Pull $\overline{OE}$ high to place 2 outputs in 3-state mode. Referenced to $V_{CCA}$. |
| GND | 8, 9 | — | Ground |
| $V_{CCA}$ | 1 | — | A-port power supply voltage. 1.2V ≤ $V_{CCA}$ ≤ 3.6V |
| $V_{CCB}$ | 16 | — | B-port power supply voltage. 1.2V ≤ $V_{CCB}$ ≤ 3.6V |

(1) I = input, O = output


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# 5 Specifications

## 5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted) (1)

| | | | MIN | MAX | UNIT |
|---|---|---|---|---|---|
| $V_{CCA}$ | Supply voltage | | –0.5 | 4.6 | V |
| $V_{CCB}$ | Supply voltage | | –0.5 | 4.6 | V |
| $V_I$ | Input voltage(2) | I/O ports (A port) | –0.5 | 4.6 | V |
| | | I/O ports (B port) | –0.5 | 4.6 | |
| | | Control inputs | –0.5 | 4.6 | |
| $V_O$ | Voltage applied to any output in the high-impedance or power-off state(2) | A port | –0.5 | 4.6 | V |
| | | B port | –0.5 | 4.6 | |
| $V_O$ | Voltage applied to any output in the high or low state(2) (3) | A port | –0.5 | $V_{CCA}$ + 0.5 | V |
| | | B port | –0.5 | $V_{CCB}$ + 0.5 | |
| $I_{IK}$ | Input clamp current | $V_I$ < 0 | | –50 | mA |
| $I_{OK}$ | Output clamp current | $V_O$ < 0 | | –50 | mA |
| $I_O$ | Continuous output current | | | ±50 | mA |
| | Continuous current through $V_{CCA}$, $V_{CCB}$, or GND | | | ±100 | mA |
| $T_{stg}$ | Storage temperature | | –65 | 150 | °C |

(1) Operation outside the *Absolute Maximum Rating* may cause permanent device damage. *Absolute Maximum Rating* do not imply functional operation of the device at these or any other conditions beyond those listed under *Recommended Operating Condition*. If used outside the *Recommended Operating Condition* but within the *Absolute Maximum Rating*, the device may not be fully functional, and this may affect device reliability, functionality, performance, and shorten the device lifetime.
(2) The input voltage and output negativeVoltage ratings may be exceeded if the input and output current ratings are observed.
(3) The output positiveVoltage rating may be exceeded up to 4.6V maximum if the output current rating is observed.

## 5.2 ESD Ratings

| | | | VALUE | UNIT |
|---|---|---|---|---|
| $V_{(ESD)}$ | Electrostatic discharge | Human body model (HBM), per ANSI/ESDA/JEDEC JS-001(1) | ±8000 | V |
| | | Charged-device model (CDM), per JEDEC specification JESD22-C101(2) | ±1000 | |
| | | Machine model | ±200 | |

(1) JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.

## 5.3 Recommended Operating Conditions

see (1) (2) (3) (4) (5)

| | | | $V_{CCI}$ | $V_{CCO}$ | MIN | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| $V_{CCA}$ | Supply voltage | | | | 1.08 | 3.6 | V |
| $V_{CCB}$ | Supply voltage | | | | 1.08 | 3.6 | V |
| | | | 1.08V | | $V_{CCI}$ × 0.7 | | |
| $V_{IH}$ | High-level input voltage | Data inputs(4) | 1.1V to 1.95V | | $V_{CCI}$ × 0.65 | | V |
| | | | 2V to 2.7V | | 1 | | |
| | | | 2.8V to 3.6V | | 1.4 | | |
| $V_{IL}$ | Low-level input voltage | Data inputs(4) | 1.08V | | | $V_{CCI}$ × 0.3 | V |
| | | | 1.1V to 1.95V | | | $V_{CCI}$ × 0.35 | |
| | | | 2V to 2.7V | | | 1.5 | |
| | | | 2V to 3.6V | | | 1.9 | |

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## 5.3 Recommended Operating Conditions (continued)

see (1) (2) (3) (4) (5)

| | | | $V_{CCI}$ | $V_{CCO}$ | MIN | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| $V_{IH}$ | High-level input voltage | DIR (referenced to $V_{CCA}$)(5) | 1.08V to 1.95V | | $V_{CCA}$ × 0.65 | | V |
| | | | 2V to 2.7V | | 1 | | |
| | | | 3V to 3.6V | | 1.3 | | |
| $V_{IL}$ | Low-level input voltage | DIR (referenced to $V_{CCA}$)(5) | 1.08V to 1.95V | | | $V_{CCA}$ × 0.35 | V |
| | | | 2V to 2.7V | | | 1.3 | |
| | | | 3V to 3.6V | | | 1.7 | |
| $V_I$ | Input voltage | | | | 0 | 3.6 | V |
| $V_O$ | Output voltage | Active state | | | 0 | $V_{CCO}$ | V |
| | | 3-state | | | 0 | 3.6 | |
| $I_{OH}$ | High-level output current | | | 1.08V to 1.32V | | –3 | mA |
| | | | | 1.4V to 1.6V | | –6 | |
| | | | | 1.65V to 1.95V | | –8 | |
| | | | | 2.3V to 2.7V | | –9 | |
| | | | | 3V to 3.6V | | –12 | |
| $I_{OL}$ | Low-level output current | | | 1.08V to 1.32V | | 3 | mA |
| | | | | 1.4V to 1.6V | | 6 | |
| | | | | 1.65V to 1.95V | | 8 | |
| | | | | 2.3V to 2.7V | | 9 | |
| | | | | 3V to 3.6V | | 12 | |
| Δt/Δv | Input transition rise or fall rate | | | | | 5 | ns/V |
| $T_A$ | Operating free-air temperature | | | | –40 | 125 | °C |

(1) $V_{CCI}$ is the $V_{CC}$ associated with the input port.
(2) $V_{CCO}$ is the $V_{CC}$ associated with the output port.
(3) All unused data inputs of the device must be held at $V_{CCI}$ or GND for proper device operation. Refer to the TI application report, *Implications of Slow or Floating CMOS Inputs*, literature number SCBA004.
(4) For $V_{CCI}$ values not specified in the data sheet, $V_{IH}$ min = $V_{CCI}$ × 0.7V, $V_{IL}$ max = $V_{CCI}$ × 0.3V.
(5) For $V_{CCA}$ values not specified in the data sheet, $V_{IH}$ min = $V_{CCA}$ × 0.7V, $V_{IL}$ max = $V_{CCA}$ × 0.3V.

## 5.4 Thermal Information

| THERMAL METRIC(1) | | SN74AVCH4T245-Q1 PW (TSSOP) 16 PINS | UNIT |
|---|---|---|---|
| $R_{\theta JA}$ | Junction-to-ambient thermal resistance(2) | 112.0 | °C/W |
| $R_{\theta JC(top)}$ | Junction-to-case (top) thermal resistance | 46.8 | °C/W |
| $R_{\theta JB}$ | Junction-to-board thermal resistance | 57.1 | °C/W |
| $\psi_{JT}$ | Junction-to-top characterization parameter | 5.7 | °C/W |
| $\psi_{JB}$ | Junction-to-board characterization parameter | 56.5 | °C/W |
| $R_{\theta JC(bot)}$ | Junction-to-case (bottom) thermal resistance | — | °C/W |

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, SPRA953.
(2) The package thermal impedance is calculated in accordance with JESD 51-7.


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## 5.5 Electrical Characteristics

All typical limits apply over $T_A$ = 25°C, and all maximum and minimum limits apply over $T_A$ = –40°C to 125°C (unless otherwise noted).(5) (6)

| PARAMETER | TEST CONDITIONS | | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|
| $V_{OH}$ | $I_{OH}$ = –100µA; $V_{CCA}$ = 1.08V to 3.6V; $V_{CCB}$ = 1.08V to 3.6V; $V_I$ = $V_{IH}$ | | $V_{CCO}$ – 0.2 | | | V |
| | $I_{OH}$ = –3mA; $V_{CCA}$ = 1.1V; $V_{CCB}$ = 1.1V; $V_I$ = $V_{IH}$ | | 0.8 | | | |
| | $I_{OH}$ = –6mA; $V_{CCA}$ = 1.4V; $V_{CCB}$ = 1.4V; $V_I$ = $V_{IH}$ | | 1.0 | | | |
| | $I_{OH}$ = –8mA; $V_{CCA}$ = 1.65V; $V_{CCB}$ = 1.65V; $V_I$ = $V_{IH}$ | | 1.2 | | | |
| | $I_{OH}$ = –9mA; $V_{CCA}$ = 2.3V; $V_{CCB}$ = 2.3V; $V_I$ = $V_{IH}$ | | 1.8 | | | |
| | $I_{OH}$ = –12mA; $V_{CCA}$ = 3V; $V_{CCB}$ = 3V; $V_I$ = $V_{IH}$ | | 2.3 | | | |
| $V_{OL}$ | $I_{OL}$ = 100µA; $V_{CCA}$ = 1.08V to 3.6V; $V_{CCB}$ = 1.08V to 3.6V; $V_I$ = $V_{IL}$ | | | | 0.2 | V |
| | $I_{OL}$ = 3mA; $V_{CCA}$ = 1.1V; $V_{CCB}$ = 1.1V; $V_I$ = $V_{IL}$ | | | | 0.2 | |
| | $I_{OL}$ = 6mA; $V_{CCA}$ = 1.4V; $V_{CCB}$ = 1.4V; $V_I$ = $V_{IL}$ | | | | 0.31 | |
| | $I_{OL}$ = 8mA; $V_{CCA}$ = 1.65V; $V_{CCB}$ = 1.65V; $V_I$ = $V_{IL}$ | | | | 0.35 | |
| | $I_{OL}$ = 9mA; $V_{CCA}$ = 2.3V; $V_{CCB}$ = 2.3V; $V_I$ = $V_{IL}$ | | | | 0.33 | |
| | $I_{OL}$ = 12mA; $V_{CCA}$ = 3V; $V_{CCB}$ = 3V; $V_I$ = $V_{IL}$ | | | | 0.40 | |
| $I_I$ DIR input | $V_I$ = $V_{CCA}$ or GND; $V_{CCA}$ = 1.08V to 3.6V; $V_{CCB}$ = 1.08V to 3.6V | $T_A$ = 25°C | -0.25 | | 0.25 | µA |
| | | $T_A$ = –40°C to 125°C | -1 | | 1.5 | |
| $I_{BHL}$ (1) | $V_I$ = 0.42V; $V_{CCA}$ = 1.08V; $V_{CCB}$ = 1.08V | | 9 | | | µA |
| | $V_I$ = 0.49V; $V_{CCA}$ = 1.4V; $V_{CCB}$ = 1.4V | | 19 | | | |
| | $V_I$ = 0.58V; $V_{CCA}$ = 1.65V; $V_{CCB}$ = 1.65V | | 29 | | | |
| | $V_I$ = 0.7V; $V_{CCA}$ = 2.3V; $V_{CCB}$ = 2.3V | | 53 | | | |
| | $V_I$ = 0.8V; $V_{CCA}$ = 3.3V; $V_{CCB}$ = 3.3V | | 86 | | | |
| $I_{BHH}$ (2) | $V_I$ = 0.78V; $V_{CCA}$ = 1.08V; $V_{CCB}$ = 1.08V | | | –25 | | µA |
| | $V_I$ = 0.91V; $V_{CCA}$ = 1.4V; $V_{CCB}$ = 1.4 | | | | -21 | |
| | $V_I$ = 1.07V; $V_{CCA}$ = 1.65V; $V_{CCB}$ = 1.65V | | | | -30 | |
| | $V_I$ = 1.6V; $V_{CCA}$ = 2.3V; $V_{CCB}$ = 2.3V | | | | -53 | |
| | $V_I$ = 2V; $V_{CCA}$ = 3.3V; $V_{CCB}$ = 3.3V | | | | -118 | |
| $I_{BHLO}$ (3) | $V_I$ = 0 to $V_{CCI}$ | $V_{CCA}$ = 1.32V; $V_{CCB}$ = 1.32V | | | 66 | µA |
| | | $V_{CCA}$ = 1.6V; $V_{CCB}$ = 1.6V | | | 103 | |
| | | $V_{CCA}$ = 1.95V; $V_{CCB}$ = 1.95V | | | 145 | |
| | | $V_{CCA}$ = 2.7V; $V_{CCB}$ = 2.7V | | | 238 | |
| | | $V_{CCA}$ = 3.6V; $V_{CCB}$ = 3.6V | | | 350 | |

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## 5.5 Electrical Characteristics (continued)

All typical limits apply over $T_A$ = 25°C, and all maximum and minimum limits apply over $T_A$ = –40°C to 125°C (unless otherwise noted).(5) (6)

| PARAMETER | | TEST CONDITIONS | | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| $I_{BHHO}$ (4) | | $V_I$ = 0 to $V_{CCI}$ | $V_{CCA}$ = 1.32V; $V_{CCB}$ = 1.32V | -48 | | | µA |
| | | | $V_{CCA}$ = 1.6V; $V_{CCB}$ = 1.6V | –80 | | | |
| | | | $V_{CCA}$ = 1.95V; $V_{CCB}$ = 1.95V | –122 | | | |
| | | | $V_{CCA}$ = 2.7V; $V_{CCB}$ = 2.7V | –218 | | | |
| | | | $V_{CCA}$ = 3.6V; $V_{CCB}$ = 3.6V | –339 | | | |
| $I_{off}$ | A port | $V_I$ or $V_O$ = 0 to 3.6V; $V_{CCA}$ = 0V; $V_{CCB}$ = 0V to 3.6V | $T_A$ = 25°C | | ±0.1 | ±1 | µA |
| | | | $T_A$ = –40°C to 125°C | | | ±5 | |
| | B port | $V_I$ or $V_O$ = 0 to 3.6V; $V_{CCA}$ = 0V to 3.6V; $V_{CCB}$ = 0V | $T_A$ = 25°C | | ±0.1 | ±1 | |
| | | | $T_A$ = –40°C to 125°C | | | ±5 | |
| $I_{OZ}$ (7) | A or B port | $V_O$ = $V_{CCO}$ or GND, $V_I$ = $V_{CCI}$ or GND; $\overline{OE}$ = $V_{IH}$; $V_{CCA}$ = 3.6V; $V_{CCB}$ = 3.6V | $T_A$ = 25°C | | ±0.5 | ±2.5 | µA |
| | | | $T_A$ = –40°C to 125°C | | | ±5 | |
| | B port | $V_O$ = $V_{CCO}$ or GND, $V_I$ = $V_{CCI}$ or GND; $\overline{OE}$ = don't care; $V_{CCA}$ = 0V; $V_{CCB}$ = 3.6V | | | | ±5 | |
| | A port | $V_O$ = $V_{CCO}$ or GND, $V_I$ = $V_{CCI}$ or GND; $\overline{OE}$ = don't care; $V_{CCA}$ = 3.6V; $V_{CCB}$ = 0V | | | | ±5 | |
| $I_{CCA}$ | | $V_I$ = $V_{CCI}$ or GND, $I_O$ = 0 | $V_{CCA}$ = 1.08V to 3.6V $V_{CCB}$ = 1.08V to 3.6V | | | 9 | µA |
| | | | $V_{CCA}$ = 0V; $V_{CCB}$ = 3.6V | | | –2 | |
| | | | $V_{CCA}$ = 3.6V; $V_{CCB}$ = 0V | | | 5 | |
| $I_{CCB}$ | | $V_I$ = $V_{CCI}$ or GND, $I_O$ = 0 | $V_{CCA}$ = 1.08V to 3.6V $V_{CCB}$ = 1.08V to 3.6V | | | 7 | µA |
| | | | $V_{CCA}$ = 0V; $V_{CCB}$ = 3.6V | | | 4.5 | |
| | | | $V_{CCA}$ = 3.6V; $V_{CCB}$ = 0V | | | –2 | |
| $I_{CCA}$ + $I_{CCB}$ | | $V_I$ = $V_{CCI}$ or GND, $I_O$ = 0; $V_{CCA}$ = 1.08V to 3.6V; $V_{CCB}$ = 1.08V to 3.6V | | | | 16 | µA |
| $C_i$ | Control inputs | $V_I$ = 3.3V or GND; $V_{CCA}$ = 3.3V; $V_{CCB}$ = 3.3V | | | | 4.5 | pF |
| $C_{io}$ | A or B port | $V_O$ = 3.3V or GND; $V_{CCA}$ = 3.3V; $V_{CCB}$ = 3.3V | | | | 5.1 | pF |

(1) The bus-hold circuit can sink at least the minimum low sustaining current at $V_{IL}$ maximum. $I_{BHL}$ should be measured after lowering $V_{IN}$ to GND and then raising it to $V_{IL}$ max.
(2) The bus-hold circuit can source at least the minimum high sustaining current at $V_{IH}$ min. $I_{BHH}$ should be measured after raising $V_{IN}$ to $V_{CC}$ and then lowering it to $V_{IH}$ min.
(3) An external driver must source at least $I_{BHLO}$ to switch this node from low to high.
(4) An external driver must sink at least $I_{BHHO}$ to switch this node from high to low.
(5) $V_{CCO}$ is the $V_{CC}$ associated with the output port.
(6) $V_{CCI}$ is the $V_{CC}$ associated with the input port.
(7) For I/O ports, the parameter $I_{OZ}$ includes the input leakage current.

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## 5.6 Switching Characteristics, $V_{CCA}$ = 1.2V ± 0.12V

over recommended operating free-air temperature range (for parameter descriptions, see Figure 6-1)

| PARAMETER | FROM (INPUT) | TO (OUTPUT) | $V_{CCB}$ | TYP | UNIT |
|---|---|---|---|---|---|
| $t_{PLH}$, $t_{PHL}$ | A | B | $V_{CCB}$ = 1.2V ± 0.12V | 3.1 | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 2.6 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 2.5 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V3 | 3 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 3.5 | |
| $t_{PLH}$, $t_{PHL}$ | B | A | $V_{CCB}$ = 1.2V ± 0.12V | 3.1 | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 2.7 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 2.5 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 2.4 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 2.3 | |
| $t_{PZH}$, $t_{PZL}$ | $\overline{OE}$ | A | $V_{CCB}$ = 1.2V ± 0.12V | 5.3 | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 5.3 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 5.3 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 5.3 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 5.3 | |
| $t_{PZH}$, $t_{PZL}$ | $\overline{OE}$ | B | $V_{CCB}$ = 1.2V ± 0.12V | 5.1 | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 4 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 3.5 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 3.2 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 3.1 | |
| $t_{PHZ}$, $t_{PLZ}$ | $\overline{OE}$ | A | $V_{CCB}$ = 1.2V ± 0.12V | 4.8 | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 4.8 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 4.8 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 4.8 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 4.8 | |
| $t_{PHZ}$, $t_{PLZ}$ | $\overline{OE}$ | B | $V_{CCB}$ = 1.2V ± 0.12V | 4.7 | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 4 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 4.1 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 4.3 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 5.1 | |

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## 5.7 Switching Characteristics, $V_{CCA}$ = 1.5V ± 0.1V

over temperature range -40 °C to +125 °C (for parameter descriptions, see Figure 6-1)

| PARAMETER | FROM (INPUT) | TO (OUTPUT) | $V_{CCB}$ | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| $t_{PHL}$, $t_{PLH}$ | A | B | $V_{CCB}$ = 1.2V ± 0.12V | | 4.2 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 2.2 | | 5.7 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 2.0 | | 4.7 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 1.7 | | 3.8 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 1.5 | | 3.4 | |
| $t_{PLH}$, $t_{PHL}$ | B | A | $V_{CCB}$ = 1.2V ± 0.12V | | 4.6 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 2.1 | | 5.7 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 1.9 | | 5.1 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 1.7 | | 4.2 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 1.6 | | 3.8 | |
| $t_{PZH}$, $t_{PZL}$ | $\overline{OE}$ | A | $V_{CCB}$ = 1.2V ± 0.12V | | 5.8 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 3.8 | | 10.6 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 3.8 | | 10.7 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 3.7 | | 10.6 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 3.7 | | 10.5 | |
| $t_{PZH}$, $t_{PZL}$ | $\overline{OE}$ | B | $V_{CCB}$ = 1.2V ± 0.12V | | 8.7 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 3.9 | | 10.8 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 3.5 | | 9.5 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 3.2 | | 8.3 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 3.1 | | 8.0 | |
| $t_{PHZ}$, $t_{PLZ}$ | $\overline{OE}$ | A | $V_{CCB}$ = 1.2V ± 0.12V | | 5.6 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 3.9 | | 9.4 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 3.9 | | 9.4 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 3.9 | | 9.4 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 3.9 | | 9.4 | |
| $t_{PHZ}$, $t_{PLZ}$ | $\overline{OE}$ | B | $V_{CCB}$ = 1.2V ± 0.12V | | 8.6 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 4.6 | | 11.0 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 4.6 | | 10.6 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 3.7 | | 8.9 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 4.2 | | 9.4 | |


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## 5.8 Switching Characteristics, $V_{CCA}$ = 1.8V ± 0.15V

over temperature range -40 °C to +125 °C (for parameter descriptions, see Figure 6-1)

| PARAMETER | FROM (INPUT) | TO (OUTPUT) | $V_{CCB}$ | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| $t_{PLH}$, $t_{PHL}$ | A | B | $V_{CCB}$ = 1.2V ± 0.12V | | 3.8 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 2.1 | | 5.1 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 2.0 | | 4.2 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 1.6 | | 3.1 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 1.4 | | 2.9 | |
| $t_{PLH}$, $t_{PHL}$ | B | A | $V_{CCB}$ = 1.2V ± 0.12V | | 4.2 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 2.2 | | 4.7 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 2.0 | | 4.2 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 1.8 | | 3.7 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 1.7 | | 3.3 | |
| $t_{PZH}$, $t_{PZL}$ | $\overline{OE}$ | A | $V_{CCB}$ = 1.2V ± 0.12V | | 4.5 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 3.4 | | 7.7 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 3.4 | | 7.7 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 3.3 | | 7.7 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 3.4 | | 7.6 | |
| $t_{PZH}$, $t_{PZL}$ | $\overline{OE}$ | B | $V_{CCB}$ = 1.2V ± 0.12V | | 8.0 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 3.9 | | 9.1 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 3.4 | | 7.9 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 3.0 | | 6.6 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 2.9 | | 6.2 | |
| $t_{PHZ}$, $t_{PLZ}$ | $\overline{OE}$ | A | $V_{CCB}$ = 1.2V ± 0.12V | | 5.3 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 4.1 | | 7.9 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 4.1 | | 8.0 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 4.1 | | 8.0 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 4.1 | | 8.0 | |
| $t_{PHZ}$, $t_{PLZ}$ | $\overline{OE}$ | B | $V_{CCB}$ = 1.2V ± 0.12V | | 7.7 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 4.5 | | 9.4 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 4.6 | | 9.1 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 3.9 | | 7.6 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 4.3 | | 8.1 | |

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## 5.9 Switching Characteristics, $V_{CCA}$ = 2.5V ± 0.2V

over temperature range -40 °C to +125 °C (for parameter descriptions, see Figure 6-1)

| PARAMETER | FROM (INPUT) | TO (OUTPUT) | $V_{CCB}$ | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| $t_{PLH}$, $t_{PHL}$ | A | B | $V_{CCB}$ = 1.2V ± 0.12V | | 3.3 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 1.9 | | 4.2 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 1.8 | | 3.7 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 1.5 | | 2.6 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 1.3 | | 2.3 | |
| $t_{PLH}$, $t_{PHL}$ | B | A | $V_{CCB}$ = 1.2V ± 0.12V | | 3.6 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 1.8 | | 3.8 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 1.6 | | 3.1 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 1.5 | | 2.6 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 1.5 | | 2.5 | |
| $t_{PZH}$, $t_{PZL}$ | $\overline{OE}$ | A | $V_{CCB}$ = 1.2V ± 0.12V | | 3.0 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 2.5 | | 4.8 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 2.5 | | 4.8 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 2.5 | | 4.8 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 2.5 | | 4.8 | |
| $t_{PZH}$, $t_{PZL}$ | $\overline{OE}$ | B | $V_{CCB}$ = 1.2V ± 0.12V | | 7.0 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 3.5 | | 7.4 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 3.1 | | 6.1 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 2.6 | | 4.9 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 2.4 | | 4.4 | |
| $t_{PHZ}$, $t_{PLZ}$ | $\overline{OE}$ | A | $V_{CCB}$ = 1.2V ± 0.12V | | 3.7 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 3.1 | | 5.3 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 3.2 | | 5.4 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 3.1 | | 5.4 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 3.1 | | 5.4 | |
| $t_{PHZ}$ | $\overline{OE}$ | B | $V_{CCB}$ = 1.2V ± 0.12V | | 4.5 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 1.5 | | 9.4 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 1.3 | | 8.2 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 1.1 | | 6.2 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 0.9 | | 5.2 | |
| $t_{PLZ}$ | $\overline{OE}$ | B | $V_{CCB}$ = 1.2V ± 0.12V | | 6.6 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 4.1 | | 7.4 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 4.2 | | 7.3 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 3.5 | | 6.0 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 4.0 | | 6.6 | |


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## 5.10 Switching Characteristics, $V_{CCA}$ = 3.3V ± 0.3V

over temperature range -40 °C to +125 °C (for parameter descriptions, see Figure 6-1)

| PARAMETER | FROM (INPUT) | TO (OUTPUT) | $V_{CCB}$ | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| $t_{PLH}$, $t_{PHL}$ | A | B | $V_{CCB}$ = 1.2V ± 0.12V | | 3.2 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 1.8 | | 3.8 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 1.7 | | 3.3 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 1.5 | | 2.5 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 1.2 | | 2.0 | |
| $t_{PLH}$, $t_{PHL}$ | B | A | $V_{CCB}$ = 1.2V ± 0.12V | | 3.4 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 1.7 | | 3.4 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 1.5 | | 2.9 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 1.3 | | 2.9 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 1.2 | | 2.0 | |
| $t_{PZH}$, $t_{PZL}$ | $\overline{OE}$ | A | $V_{CCB}$ = 1.2V ± 0.12V | | 2.4 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 2.2 | | 3.6 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 2.2 | | 3.6 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 2.2 | | 3.6 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 2.2 | | 3.6 | |
| $t_{PZH}$, $t_{PZL}$ | $\overline{OE}$ | B | $V_{CCB}$ = 1.2V ± 0.12V | | 6.7 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 3.2 | | 6.7 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 2.8 | | 5.4 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 2.4 | | 4.2 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 2.2 | | 3.7 | |
| $t_{PHZ}$, $t_{PLZ}$ | $\overline{OE}$ | A | $V_{CCB}$ = 1.2V ± 0.12V | | 4.0 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 3.5 | | 5.5 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 3.5 | | 5.5 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 3.4 | | 5.4 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 3.5 | | 5.4 | |
| $t_{PHZ}$, $t_{PLZ}$ | $\overline{OE}$ | B | $V_{CCB}$ = 1.2V ± 0.12V | | 6.3 | | ns |
| | | | $V_{CCB}$ = 1.5V ± 0.1V | 4.0 | | 6.6 | |
| | | | $V_{CCB}$ = 1.8V ± 0.15V | 4.0 | | 6.5 | |
| | | | $V_{CCB}$ = 2.5V ± 0.2V | 3.3 | | 5.3 | |
| | | | $V_{CCB}$ = 3.3V ± 0.3V | 3.7 | | 5.9 | |

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## 5.11 Operating Characteristics

$T_A$ = 25°C

| PARAMETER | | | TEST CONDITIONS | $V_{CCA}$ | TYP | UNIT |
|---|---|---|---|---|---|---|
| $C_{pdA}$ (1) | A to B | Outputs enabled | $C_L$ = 0, f = 10MHz, $t_r$ = $t_f$ = 1ns | $V_{CCA}$ = $V_{CCB}$ = 1.2V | 1 | pF |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.5V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.8V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 2.5V | 1.5 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 3.3V | 2 | |
| | | Outputs disabled | $C_L$ = 0, f = 10MHz, $t_r$ = $t_f$ = 1ns | $V_{CCA}$ = $V_{CCB}$ = 1.2V | 1 | pF |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.5V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.8V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 2.5V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 3.3V | 1 | |
| | B to A | Outputs enabled | $C_L$ = 0, f = 10MHz, $t_r$ = $t_f$ = 1ns | $V_{CCA}$ = $V_{CCB}$ = 1.2V | 12 | pF |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.5V | 12.5 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.8V | 13 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 2.5V | 14 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 3.3V | 15 | |
| | | Outputs disabled | $C_L$ = 0, f = 10MHz, $t_r$ = $t_f$ = 1ns | $V_{CCA}$ = $V_{CCB}$ = 1.2V | 1 | pF |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.5V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.8V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 2.5V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 3.3V | 1 | |
| $C_{pdB}$ (1) | A to B | Outputs enabled | $C_L$ = 0, f = 10MHz, $t_r$ = $t_f$ = 1ns | $V_{CCA}$ = $V_{CCB}$ = 1.2V | 12 | pF |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.5V | 12.5 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.8V | 13 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 2.5V | 14 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 3.3V | 15 | |
| | | Outputs disabled | $C_L$ = 0, f = 10MHz, $t_r$ = $t_f$ = 1ns | $V_{CCA}$ = $V_{CCB}$ = 1.2V | 1 | pF |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.5V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.8V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 2.5V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 3.3V | 1 | |
| | B to A | Outputs enabled | $C_L$ = 0, f = 10MHz, $t_r$ = $t_f$ = 1ns | $V_{CCA}$ = $V_{CCB}$ = 1.2V | 1 | pF |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.5V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.8V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 2.5V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 3.3V | 2 | |
| | | Outputs disabled | $C_L$ = 0, f = 10MHz, $t_r$ = $t_f$ = 1ns | $V_{CCA}$ = $V_{CCB}$ = 1.2V | 1 | pF |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.5V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 1.8V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 2.5V | 1 | |
| | | | | $V_{CCA}$ = $V_{CCB}$ = 3.3V | 1 | |

(1) Power dissipation capacitance per transceiver. Refer to TI application report, CMOS Power Consumption and Cpd Calculation (SCAA035)


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## 5.12 Typical Characteristics

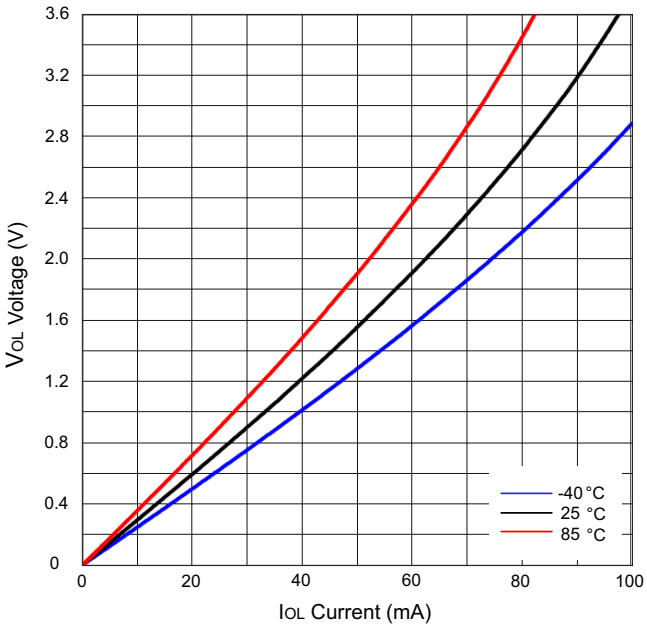



**Figure 5-1. Low-Level Output Voltage ($V_{OL}$) vs Low-Level Current ($I_{OL}$)**

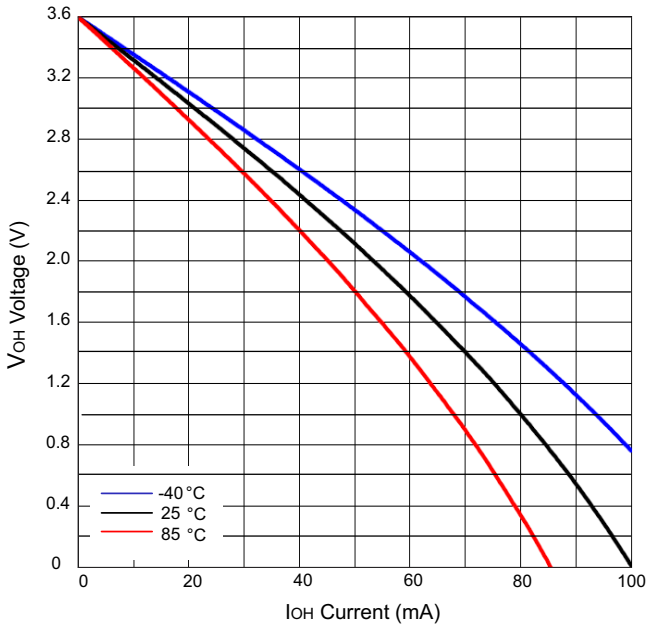
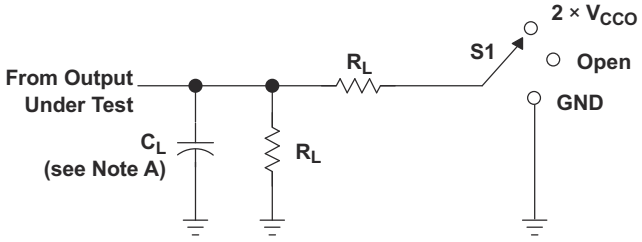



**Figure 5-2. High-Level Output Voltage ($V_{OH}$) vs High-Level Current ($I_{OH}$)**

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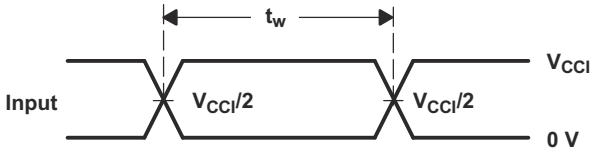
## 6 Parameter Measurement Information



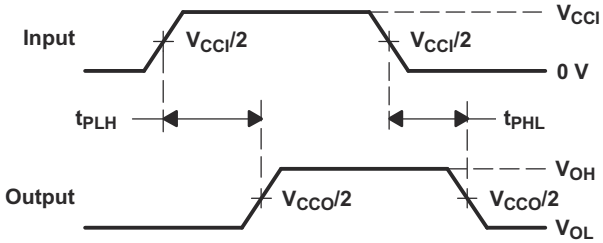

LOAD CIRCUIT

| TEST | S1 |
|---|---|
| $t_{pd}$ | Open |
| $t_{PLZ}/t_{PZL}$ | 2 × $V_{CCO}$ |
| $t_{PHZ}/t_{PZH}$ | GND |

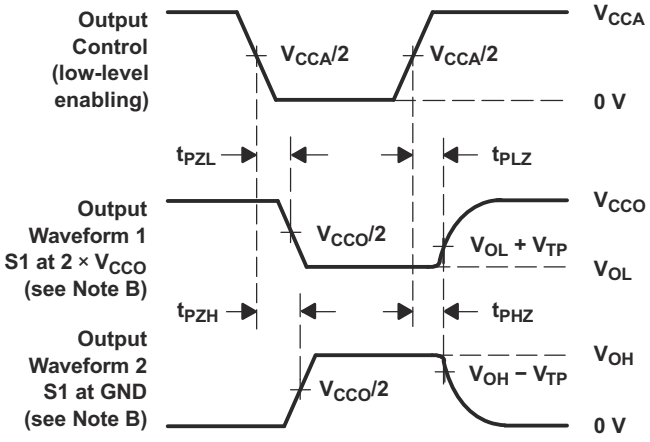
| $V_{CCO}$ | $C_L$ | $R_L$ | $V_{TP}$ |
|---|---|---|---|
| 1.2 V | 15 pF | 2 kΩ | 0.1 V |
| 1.5 V ± 0.1 V | 15 pF | 2 kΩ | 0.1 V |
| 1.8 V ± 0.15 V | 15 pF | 2 kΩ | 0.15 V |
| 2.5 V ± 0.2 V | 15 pF | 2 kΩ | 0.15 V |
| 3.3 V ± 0.3 V | 15 pF | 2 kΩ | 0.3 V |




VOLTAGE WAVEFORMS
PULSE DURATION




VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES




VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES

NOTES:
A. $C_L$ includes probe and jig capacitance.
B. Waveform 1 is for an output with internal conditions such that the output is low, except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high, except when disabled by the output control.
C. All input pulses are supplied by generators having the following characteristics: PRR 10 MHz, $Z_O$ = 50 Ω, dv/dt ≥ 1 V/ns, dv/dt ≥1 V/ns.
D. The outputs are measured one at a time, with one transition per measurement.
E. $t_{PLZ}$ and $t_{PHZ}$ are the same as $t_{dis}$.
F. $t_{PZL}$ and $t_{PZH}$ are the same as $t_{en}$.
G. $t_{PLH}$ and $t_{PHL}$ are the same as $t_{pd}$.
H. $V_{CCI}$ is the $V_{CC}$ associated with the input port.
I. $V_{CCO}$ is the $V_{CC}$ associated with the output port.

**Figure 6-1. Load Circuit and Voltage Waveforms**


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# 7 Detailed Description

## 7.1 Overview

The SN74AVCH4T245-Q1 is a 4-bit, dual-supply noninverting bidirectional voltage level translation device. Ax pins and control pins (1DIR, 2DIR,1 $\overline{OE}$, and 2 $\overline{OE}$) are supported by $V_{CCA}$, and Bx pins are supported by $V_{CCB}$. The A port can accept I/O voltages ranging from 1.08V to 3.6V, while the B port can accept I/O voltages from 1.08V to 3.6V. A high on DIR allows data transmission from Ax to Bx and a low on DIR allows data transmission from Bx to Ax when $\overline{OE}$ is set to low. When $\overline{OE}$ is set to high, both Ax and Bx pins are in the high-impedance state. For more information, refer to the *AVC Logic Family Technology and Applications* application report.

## 7.2 Functional Block Diagram

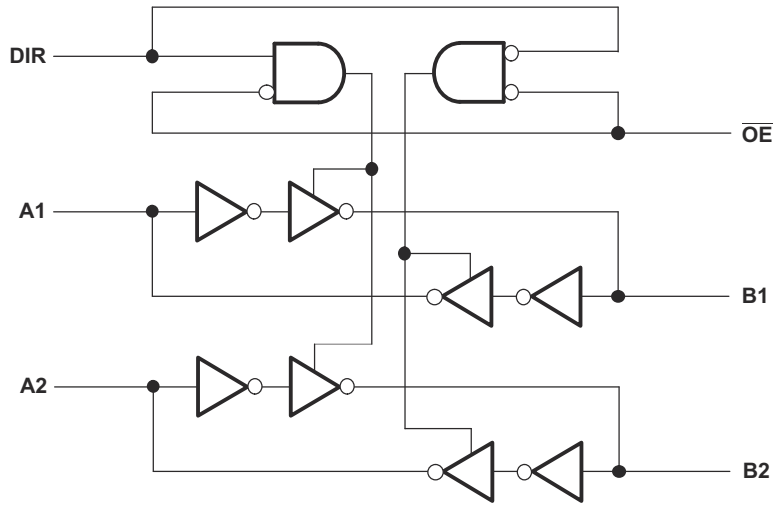



**Figure 7-1. Logic Diagram (Positive Logic) for 1/2 of SN74AVCH4T245-Q1**

## 7.3 Feature Description

### 7.3.1 Fully Configurable Dual-Rail Design

Fully configurable dual-rail design allows each port to operate over the full 1.08V to 3.6V power-supply range.

Both $V_{CCA}$ and $V_{CCB}$ can be supplied at any voltage between 1.08V and 3.6V; thus, making the device an excellent choice for translating between any of the low voltage nodes (1.2V, 1.8V, 2.5V, and 3.3V).

### 7.3.2 Supports High Speed Translation

The SN74AVCH4T245-Q1 device can support high data rate applications. The translated signal data rate can be up to 500Mbps when the signal is translated from 1.08V to 3.3V.

### 7.3.3 $I_{off}$ Supports Partial-Power-Down Mode Operation

$I_{off}$ will prevent backflow current by disabling I/O output circuits when device is in partial-power-down mode.

### 7.3.4 Bus-Hold Circuitry

This device has active bus-hold circuitry that holds unused or undriven inputs at a valid logic state. Use of pull-up or pull-down resistors with the bus-hold circuitry is not recommended. (Refer to the *Bus-Hold Circuit* application report. Pullup and pulldown resistors are not recommended on the inputs of devices with bus-hold. Unused inputs can be left floating.

### 7.3.5 Vcc Isolation Feature

The VCC isolation feature is designed so that if either $V_{CCA}$ or $V_{CCB}$ are at GND (or < 0.4V), both ports will be in a high-impedance state (IOZ shown in *Section 5.5*). This prevents false logic levels from being presented to either bus.

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## 7.4 Device Functional Modes

Table 7-1 lists the functional modes of the SN74AVCH4T245-Q1.

**Table 7-1. Function Table (Each 2-Bit Section)**

| CONTROL INPUTS(1) | | OUTPUT CIRCUITS | | OPERATION |
|---|---|---|---|---|
| $\overline{OE}$ | DIR | A PORT | B PORT | |
| L | L | Enabled | Hi-Z | B data to A bus |
| L | H | Hi-Z | Enabled | A data to B bus |
| H | X | Hi-Z | Hi-Z | Isolation |

(1) Input circuits of the data I/Os are always active.

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# Application and Implementation

**Note**

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

## 8.1 Application Information

The SN74AVCH4T245-Q1 device can be used in level-shifting applications for interfacing devices or systems operating at different interface voltages with one another. The SN74AVCH4T245-Q1 device is an excellent choice for applications where a push-pull driver is connected to the data I/Os. The maximum data rate can be up to 500Mbps when device translates a signal from 1.08V to 3.3V.

## 8.2 Typical Application

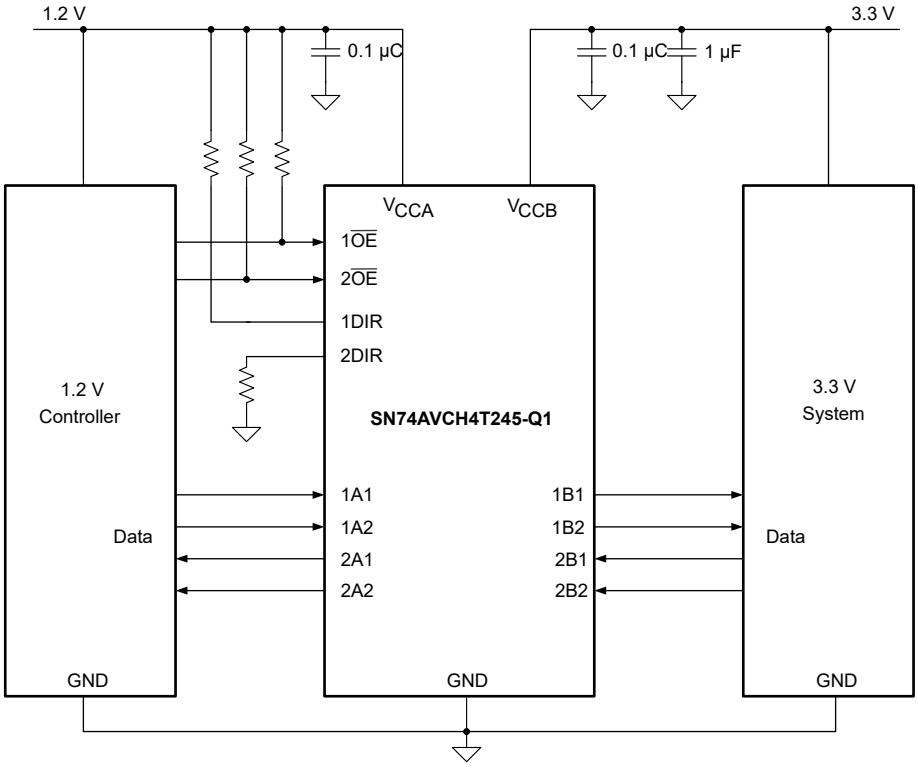



**Figure 8-1. Typical Application Diagram**

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### 8.2.1 Design Requirements

For the design example shown in Section 8.2 use the parameters listed in Table 8-1.

**Table 8-1. Design Parameters**

| DESIGN PARAMETER | EXAMPLE VALUE |
|---|---|
| Input voltage range | 1.08V to 3.6V |
| Output voltage range | 1.08V to 3.6V |

### 8.2.2 Detailed Design Procedure

To begin the design process, determine the following:

- Input voltage range
  - Use the supply voltage of the device that is driving the SN74AVCH4T245-Q1 device to determine the input voltage range. For a valid logic high, the value must exceed the $V_{IH}$ of the input port. For a valid logic low, the value must be less than the $V_{IL}$ of the input port.
- Output voltage range
  - Use the supply voltage of the device that the SN74AVCH4T245-Q1 device is driving to determine the output voltage range.

### 8.2.3 Application Curve

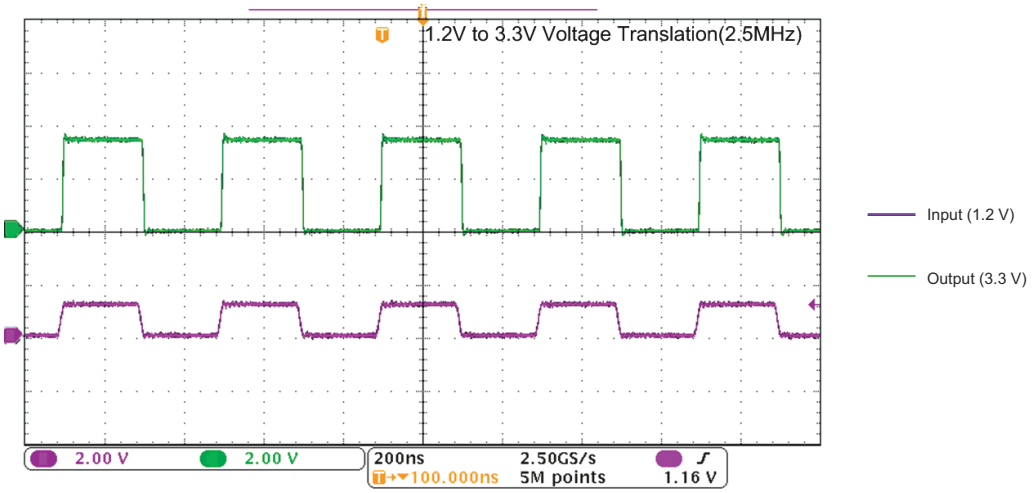



**Figure 8-2. Translation Up (1.2V to 3.3V) at 2.5MHz**

## 8.3 Power Supply Recommendations

The SN74AVCH4T245-Q1 device uses two separate configurable power-supply rails, $V_{CCA}$ and $V_{CCB}$. $V_{CCA}$ accepts any supply voltage from 1.08V to 3.6V, and $V_{CCB}$ accepts any supply voltage from 1.08V to 3.6V. The A port and B port are designed to track $V_{CCA}$ and $V_{CCB}$ respectively allowing for low voltage bidirectional translation between any of the 1.2-V, 1.5-V, 1.8-V, 2.5-V, and 3.3-V voltage nodes.

The output-enable ($\overline{OE}$) input circuit is designed so that it is supplied by $V_{CCA}$, and all outputs are placed in the high-impedance state when the $\overline{OE}$ input is high. To put the outputs in the high-impedance state during power up or power down, the $\overline{OE}$ input pin must be tied to $V_{CCA}$ through a pull-up resistor and must not be enabled until $V_{CCA}$ and $V_{CCB}$ are fully ramped and stable. The current-sinking capability of the driver determines the minimum value of the pull-up resistor to $V_{CCA}$.

$V_{CCA}$ or $V_{CCB}$ can be powered up first. If the SN74AVCH4T245-Q1 is powered up in a permanently enabled state, pull-up resistors are recommended at the input. This allows for proper or glitch-free power-up. For more information, refer to *Designing with SN74LVCXT245 and SN74LVCHXT245 Family of Direction Controlled Voltage Translators/Level-Shifters* application note.


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## 8.4 Layout

### 8.4.1 Layout Guidelines

For device reliability, it is recommended to follow common printed-circuit board layout guidelines such as:

- Use bypass capacitors on power supplies.
- Use short trace lengths to avoid excessive loading.
- Place pads on the signal paths for loading capacitors or pull-up resistors to help adjust rise and fall times of signals, depending on the system requirements.

### 8.4.2 Layout Example

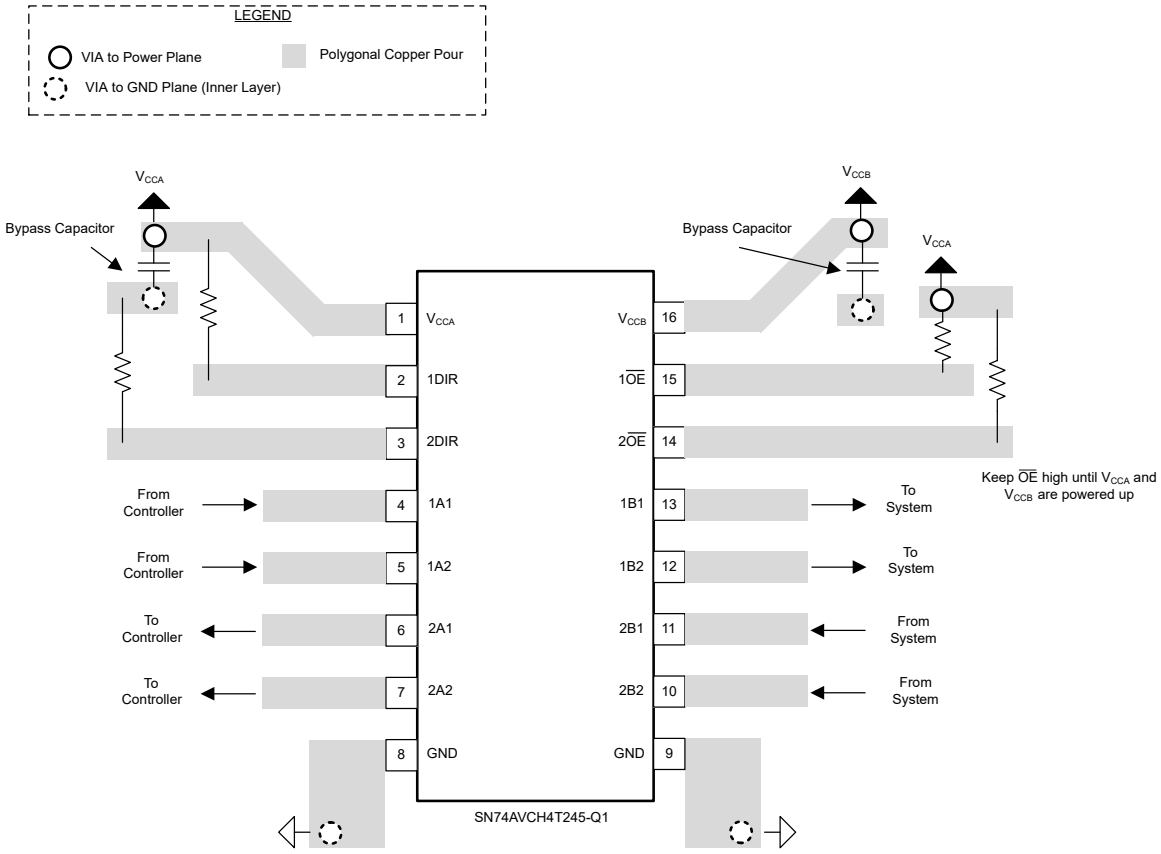



**Figure 8-3. Layout Recommendation**

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ADVANCE INFORMATION

# 8 Device and Documentation Support

## 8.1 Documentation Support

### 8.1.1 Related Documentation

For related documentation, see the following:

- Texas Instruments, *Designing with SN74LVCXT245 and SN74LVCHXT245 Family of Direction Controlled Voltage Translators/Level-Shifters*
- Texas Instruments, *Bus-Hold Circuit*
- Texas Instruments, *AVC Logic Family Technology and Applications*

## 8.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

## 8.3 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

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## 8.4 Trademarks

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## 8.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

## 8.6 Glossary

TI Glossary — This glossary lists and explains terms, acronyms, and definitions.

# 9 Revision History

| DATE | REVISION | NOTES |
|---|---|---|
| February 2024 | * | Initial Release |

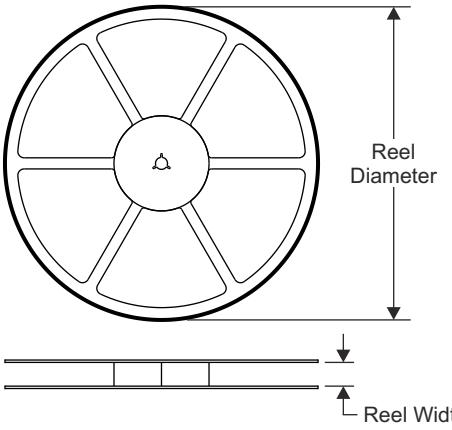
# 10 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

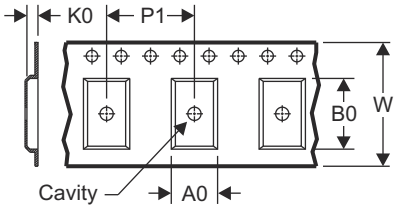

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## 10.1 Tape and Reel Information

**REEL DIMENSIONS**

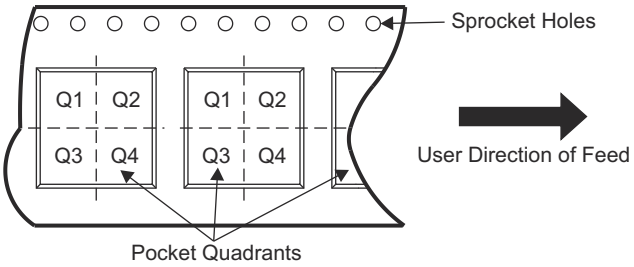



**TAPE DIMENSIONS**




| | |
|---|---|
| A0 | Dimension designed to accommodate the component width |
| B0 | Dimension designed to accommodate the component length |
| K0 | Dimension designed to accommodate the component thickness |
| W | Overall width of the carrier tape |
| P1 | Pitch between successive cavity centers |

**QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE**



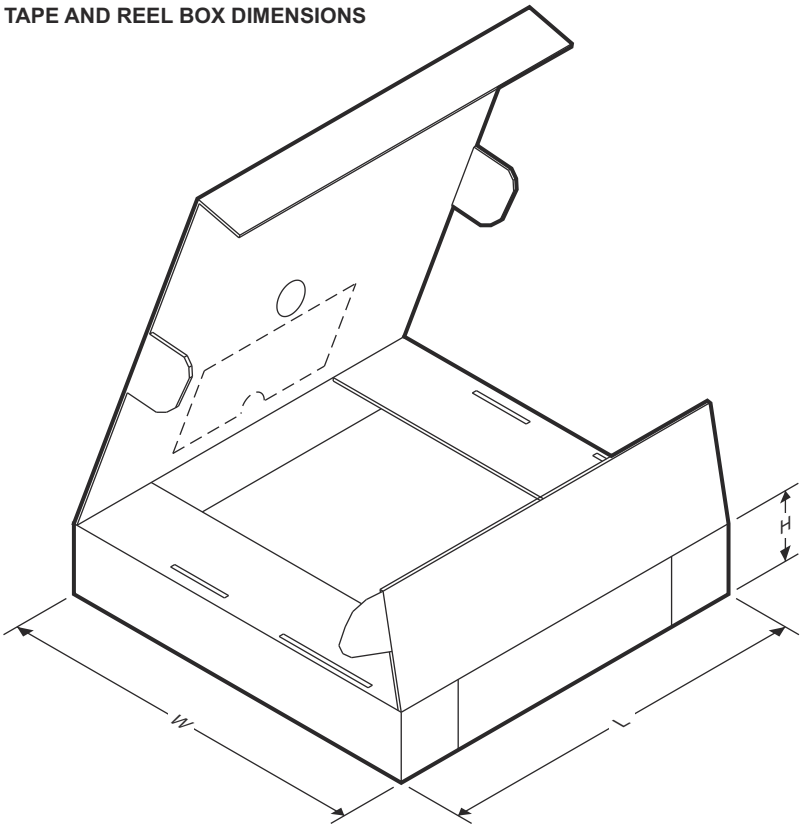

| Device | Package Type | Package Drawing | Pins | SPQ | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SN74AVC4T245PWRQ1 | TSSOP | PW | 16 | 2000 | 330.0 | 12.4 | 6.9 | 5.6 | 1.6 | 8.0 | 12.0 | Q1 |
| SN74AVC4T245PWTQ1 | TSSOP | PW | 16 | 250 | 330.0 | 12.4 | 6.9 | 5.6 | 1.6 | 8.0 | 12.0 | Q1 |

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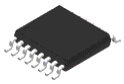
**TAPE AND REEL BOX DIMENSIONS**




| Device | Package Type | Package Drawing | Pins | SPQ | Length (mm) | Width (mm) | Height (mm) |
|---|---|---|---|---|---|---|---|
| SN74AVC4T245PWRQ1 | TSSOP | PW | 16 | 2000 | 356.0 | 356.0 | 35.0 |
| SN74AVC4T245PWTQ1 | TSSOP | PW | 16 | 250 | 356.0 | 356.0 | 35.0 |


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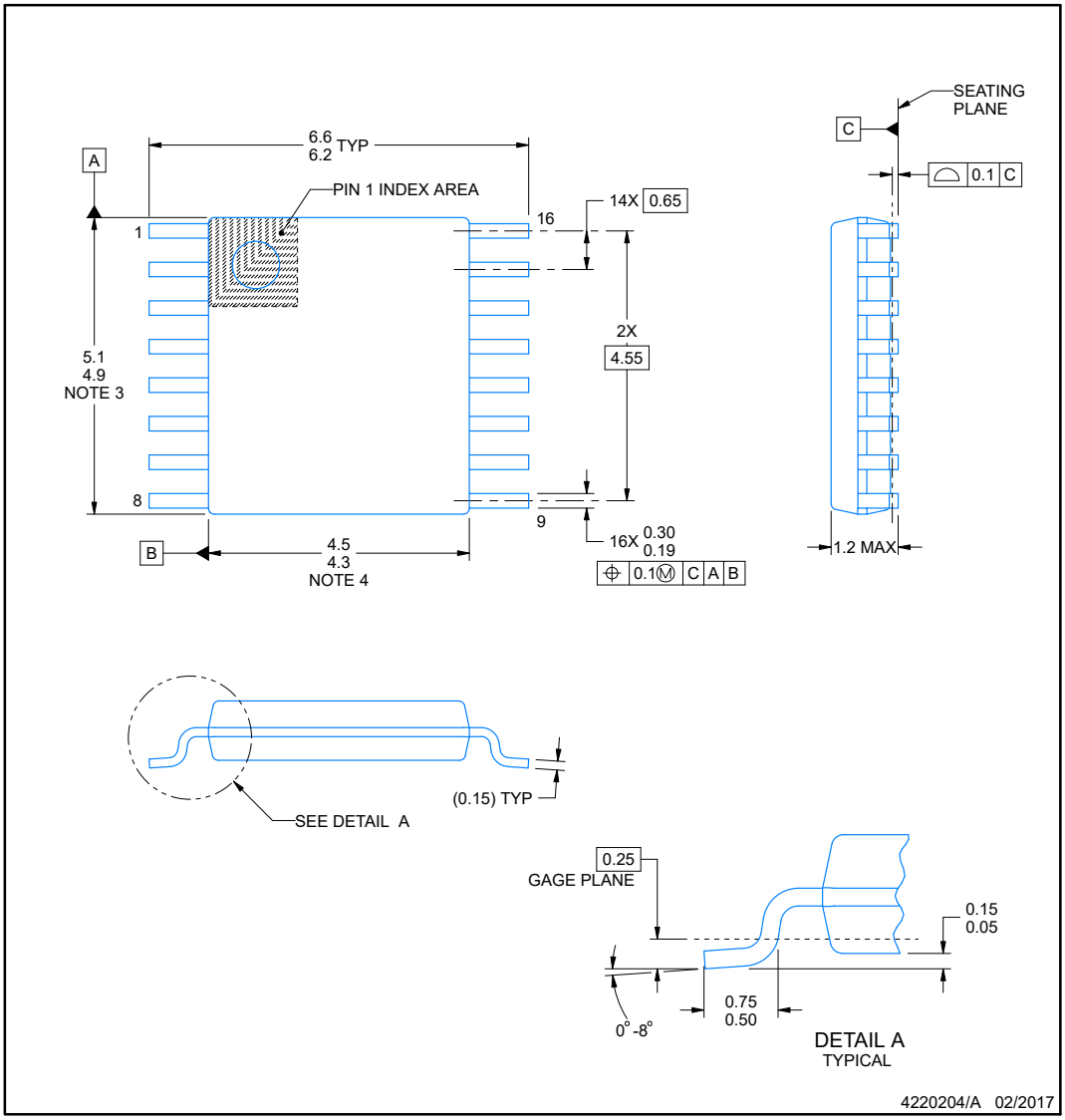
## 10.2 Mechanical Data



**PACKAGE OUTLINE**

**PW0016A** **TSSOP - 1.2 mm max height**

SMALL OUTLINE PACKAGE




NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.



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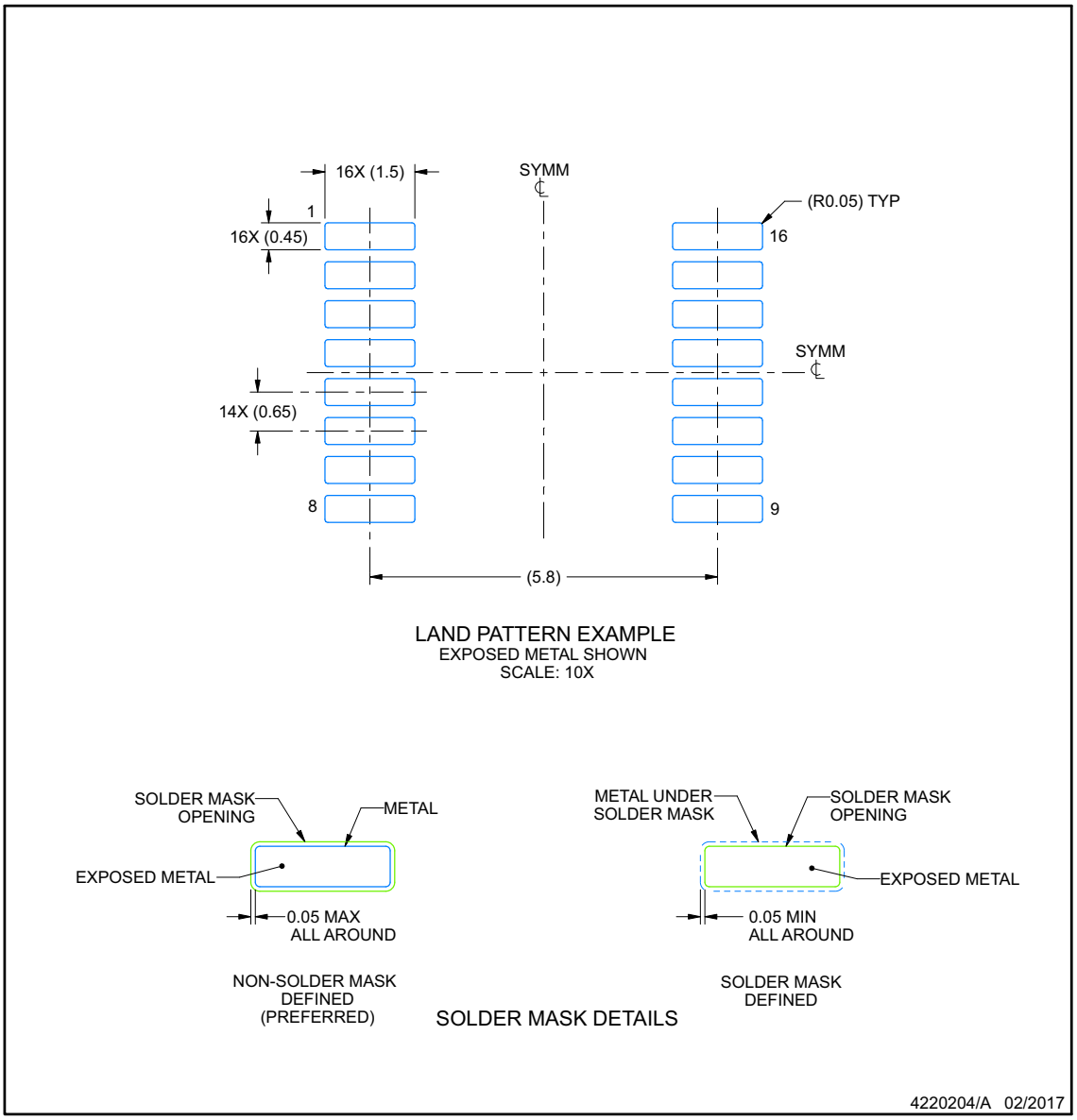
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## EXAMPLE BOARD LAYOUT

### PW0016A

### TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE




NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

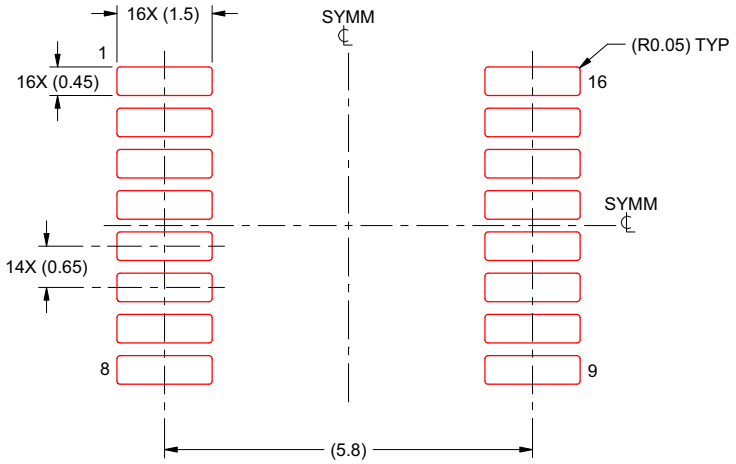

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# EXAMPLE STENCIL DESIGN

## PW0016A

### TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE




SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

4220204/A 02/2017

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.



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## PACKAGING INFORMATION

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| 74AVCH4T245QPWRQ1 | Active | Production | TSSOP (PW) \| 16 | 3000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | WS245Q |
| 74AVCH4T245QPWRQ1.A | Active | Production | TSSOP (PW) \| 16 | 3000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | WS245Q |

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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**OTHER QUALIFIED VERSIONS OF SN74AVCH4T245-Q1 :**

- Catalog : SN74AVCH4T245

- Enhanced Product : SN74AVCH4T245-EP

NOTE: Qualified Version Definitions:

- Catalog - TI's standard catalog product
- Enhanced Product - Supports Defense, Aerospace and Medical Applications

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